

```

model_name = "Helsinki-NLP/opus-mt-en-fr"
tokenizer = MarianTokenizer.from_pretrained(model_name)
model = MarianMTModel.from_pretrained(model_name)
# Translate text
text = "Hello, how are you?"
encoded_text = tokenizer(text, return_tensors="pt",
padding=True, truncation=True)
translated = model.generate(**encoded_text)
output = tokenizer.decode(translated[0], skip_special_
tokens=True)
print("Translated Text:", output)

```

Open source AI provides an excellent starting point for beginners to learn, experiment, and contribute to the growing AI community. By leveraging platforms like GitHub and Kaggle, learners can access real-world projects, collaborate with experts, and enhance their AI proficiency. The next step is to explore these projects,

Application	Tool used	Key benefit
Chatbot	Rasa	Automates customer support and FAQs
Speech-to-text	OpenWhisper	Converts audio to text for transcription and accessibility
Resume screening	NLP	Automates candidate shortlisting for HR
Medical diagnosis	TensorFlow (CNN)	AI-assisted disease detection from images
Language translation	MarianMT	Provides real-time multilingual translations

experiment with AI tools, and actively participate in the open source AI movement. 

References

- <https://teachablemachine.withgoogle.com/>
- <https://huggingface.co/docs/transformers/index>
- <https://opencv.org/>
- <https://www.fast.ai/>
- <https://ludwig.ai/>

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Robot Framework: Robot Framework is an open source automation framework that supports various testing types, including API testing. While it does not have built-in AI capabilities, it can be extended with AI libraries and tools to enhance its testing capabilities. The flexibility of Robot Framework allows teams to customise their testing approach based on their specific needs.

Benefits of AI-powered API testing

The integration of AI into API testing offers numerous advantages.

Increased efficiency: Automation of repetitive tasks reduces manual effort, allowing testers to focus on more complex scenarios.

Improved accuracy: AI algorithms minimise human error, leading to more reliable test results.

Faster time-to-market:

With accelerated testing cycles, organisations can release updates and new features more quickly, staying competitive in the market.

Cost savings: By reducing the need for extensive manual testing and identifying issues early, organisations can save on development and maintenance costs.

Challenges and considerations

While AI-powered API testing presents significant benefits, it is not without challenges. Organisations must consider the following:

Data quality: The effectiveness of AI algorithms relies on high-quality data. Poor data can lead to inaccurate predictions and test results.

Integration: Implementing AI tools requires careful integration with existing testing frameworks and workflows.

Skill gap: Teams may need training to effectively utilise AI-powered testing tools and interpret their results.

As technology continues to evolve, embracing AI in API testing will be essential for businesses looking to stay ahead in the competitive software development arena. With the rise of open source tools and current trends in testing practices, the future of API testing looks promising, paving the way for more efficient and effective software development processes. 

References

- <https://testsigma.com/>
- <https://www.zartis.com/>

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